

10 Get the odd letters

Next you need to create a similar function to produce a list of all the odd-numbered letters in your message. Put this function under the code in Step 9.

```
def get_odd_letters(message):
    odd_letters = []
    for counter in range(0, len(message)):
        if not is_even(counter):
            odd_letters.append(message[counter])
    return odd_letters
```

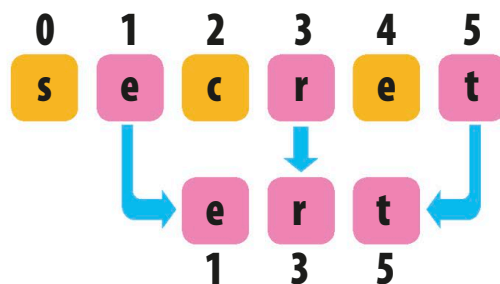
11 Swap the letters round

Now that you've got even letters in one list and odd in another, you can use them to encrypt your message. The next function will take letters alternately from these lists and put them into a new list. But rather than assembling them in the original order, starting with an even letter, it'll start the message with an odd one. Type this function under the code in Step 10.

```
def swap_letters(message):
    letter_list = []
    if not is_even(len(message)):
        message = message + 'x'
    even_letters = get_even_letters(message)
    odd_letters = get_odd_letters(message)
    for counter in range(0, int(len(message)/2)):
        letter_list.append(odd_letters[counter])
        letter_list.append(even_letters[counter])
    new_message = ''.join(letter_list)
    return new_message
```

▷ How it works

The `swap_letters()` function puts all the odd and even numbers into a new list, adding them alternately. It starts the list with the second letter in the word, which Python counts as an odd number.

**REMEMBER****Lists and length**

Python counts from 0 in lists and strings, and uses the function `len()` to find the length of a string. For example, if you type `len('secret')`, Python will tell you that the string `'secret'` is six characters long. But because the first letter is in position 0, the last letter is in position 5, not 6.

Add an extra x to any message with an odd number of letters.

Loop through the lists of odd and even letters.

Add the next odd letter to the final message.

Add the next even letter to the final message.

The `join()` function turns the list of letters into a string.

